User Requirement Analysis Report

AU Roadmap Project — Adelaide University

Project Name: Adelaide University Program Roadmap

Course: Industry Research Project

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1. Introduction

1.1 Purpose of the Document

The AU Roadmap project requires a clear, structured understanding of its users before design or implementation decisions are made. This user requirement analysis defines the needs of the main user groups, explains the problems the system is intended to solve, and translates those needs into a structured set of functional and non-functional requirements.

The document supports later project deliverables, including the information architecture design, technical documentation, implementation roadmap, prototype evaluation, and final report. It also provides a clear justification for why specific features were included in the prototype and how those features relate to real user goals.

1.2 Project Background

The Adelaide University Program Roadmap project was developed as part of the Industry Research Project. As defined in the original Project Proposal [5], the platform follows a two-level structure organising content across two levels: Level 1 is an assistance environment for current students, providing access to information about their studies, program, industry connections, alumni, and support resources; Level 2 is a public-facing website that acts as a marketing and information tool for prospective students.

The central idea behind the project is that program-related information is often fragmented. Students may need to check separate systems or webpages to understand their roadmap, timetable, support services, and future career opportunities. Prospective students may also struggle to compare programs, understand career outcomes, or navigate application information when content is spread across different pages and formats. The AU Roadmap platform therefore attempts to bring this information into a single structured system with different information environments for different user groups.

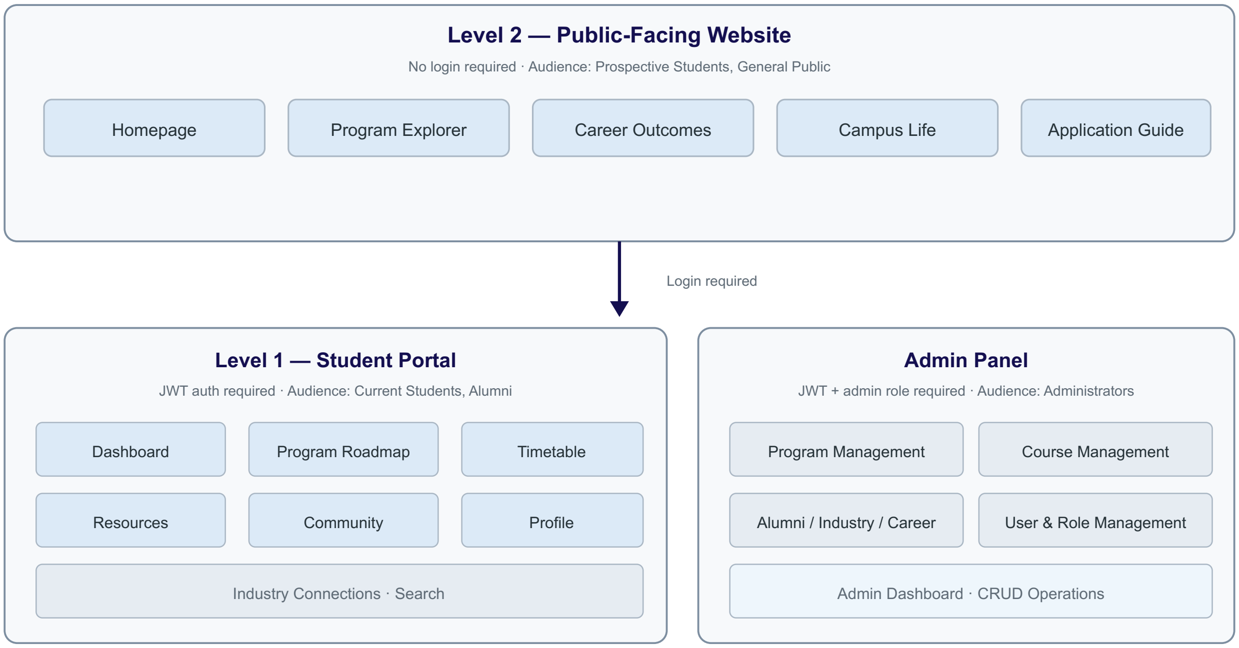

Figure 1. AU Roadmap two-level platform architecture.

1.3 Scope of this Report

The scope is user and stakeholder requirements, not implementation detail. The analysis identifies who the users are, what problems they experience, what they need the system to do, what quality expectations the system must meet, and what assumptions and constraints shaped the project. It defines the "what" and the "why" of the project, rather than the "how".

1.4 Intended Audience and Document Use

Several distinct audiences rely on this report. The project team uses it as the central reference for ensuring that design and implementation decisions remain grounded in user needs. The academic mentor and university evaluators use it to confirm that the prototype reflects a coherent, user-centred process. Future developers or contributors who pick up the project beyond its current phase use it as the entry point for understanding why each major feature exists and which user goals it serves. Finally, the client representative and stakeholder community can use this document to validate that the requirements correctly reflect the agreed-upon project scope.

1.5 Definitions, Acronyms, and Abbreviations

The following terms are used throughout this document with the meanings listed below.

Term / Acronym	Definition
AU Roadmap	The Adelaide University Program Roadmap platform — the system being specified by this report.

URS	User Requirements Specification — this document type.
FR	Functional Requirement — a system behaviour requirement, prefixed FR-XX in this report.
NFR	Non-Functional Requirement — a quality or constraint requirement, prefixed NFR-XX in this report.
Level 1 / Level 2	The two-tier platform architecture. Level 1 is the authenticated student portal; Level 2 is the public-facing exploration site.
CRUD	Create, Read, Update, Delete — standard data management operations supported by admin functions.
JWT	JSON Web Token — token-based authentication mechanism used by the platform's protected APIs.
REST / JSON	Architectural style and data format used for the platform's HTTP API responses.
WCAG	Web Content Accessibility Guidelines — the international standard referenced for accessibility expectations.
CAPTCHA	Completely Automated Public Turing test — used by the platform's registration and login (Cloudflare Turnstile) to deter automated abuse.
SRS / StRS	Software Requirements Specification / Stakeholder Requirements Specification, as defined by ISO/IEC/IEEE 29148.

Table 1. Definitions, acronyms, and abbreviations used in this document.

1.6 Document Overview

The document follows a three-part structure. The first part (Sections 1–3) provides the introduction, analysis approach, and problem context. The second part (Sections 4–8) describes who the users are, what they need, and how their needs are expressed as structured requirement statements. The third part (Sections 9–14) presents the formal functional, non-functional, and interface requirements, along with prioritisation, constraints, traceability, and evaluation against project goals. A requirement summary and appendix complete the document.

1.7 Relationship to Standards and Best Practice

The structure and terminology of this report draw on established requirements engineering standards, most notably ISO/IEC/IEEE 29148:2018 (Requirements engineering) and ISO/IEC 25065:2019 (Common Industry Format for Usability: User Requirements Specification). The report adopts the stakeholder identification, user profiling, and structured functional/non-functional requirement patterns recommended by these standards, This report follows the structure recommended by ISO/IEC 25065:2019 for user requirements specification [1]. It also aligns with ISO/IEC/IEEE 29148:2018 [2], which defines the life-cycle processes for requirements engineering. The report adopts the stakeholder identification, user profiling, and structured functional/non-functional requirement patterns recommended by these standards, adapted to the scale and context of a university course project.

2. Requirement Analysis Approach

2.1 Requirement Elicitation Method

Requirements were gathered through a multi-source approach: analysis of the project proposal and scope documentation, review of the early project concept and platform architecture, walkthrough of initial prototype pages, and iterative feedback from the academic mentor. The user-centred approach reflects the principles of ISO 9241-210:2019 [3]. This approach reflects the course project context, where the client and developer roles overlap and the requirement baseline emerges progressively through prototyping and refinement rather than a standalone discovery phase.

2.2 Analysis Principles

Three principles guided the requirement analysis. First, user-centred: the analysis starts from who the users are and what they need, not from technical features. Second, traceable: each requirement is assigned an identifier and can be mapped to prototype pages and later evaluation criteria. Third, prioritised: requirements are ranked as High, Medium, or Low to guide implementation effort and scope decisions within the project timeline.

2.3 Deliverable Context

This user requirement analysis report serves as the upstream reference for subsequent project deliverables. The Information Architecture Design translates requirements into page and navigation structures. The Technical Documentation references requirement IDs when describing backend controllers, API endpoints, and database schemas. The Implementation Roadmap maps requirements to development phases. The System Testing and Evaluation Report uses requirements as verification criteria. The Final Report summarises the requirement-to-deliverable traceability.

3. Project Context and Problem Definition

3.1 Context of Use

The AU Roadmap platform exists within the Adelaide University digital ecosystem. Students currently access program information, timetables, support resources, career content, and community discussion across multiple websites and systems. These systems are not always structured around how students think about their academic journey. The platform aims to consolidate key information and present it in a task-oriented way that aligns with student planning, exploration, and decision-making needs.

3.2 Problem Statement

Program-related information at the university is fragmented. A current student who wants to understand degree progression must consult the program handbook, a separate timetable system, and multiple resource and career pages. A prospective student who wants to compare programs must navigate between multiple entry points. University administrators who want to maintain this information must update content across disconnected systems. The AU Roadmap platform addresses this fragmentation by providing a single structured environment organised around the user's primary goal: understanding and planning their study path.

3.3 User Problem Summary

User Group	Core Problem	Consequence
Current Students	Fragmented academic planning and support information	Slower decision-making; difficulty connecting study choices to career goals
Prospective Students	Dispersed program and outcome information	Uncertainty in comparison and enrolment decisions
Administrators	Inconsistent content across domains	Higher maintenance effort; risk of contradictory information

Table 2. Summary of core user problems addressed by the AU Roadmap platform.

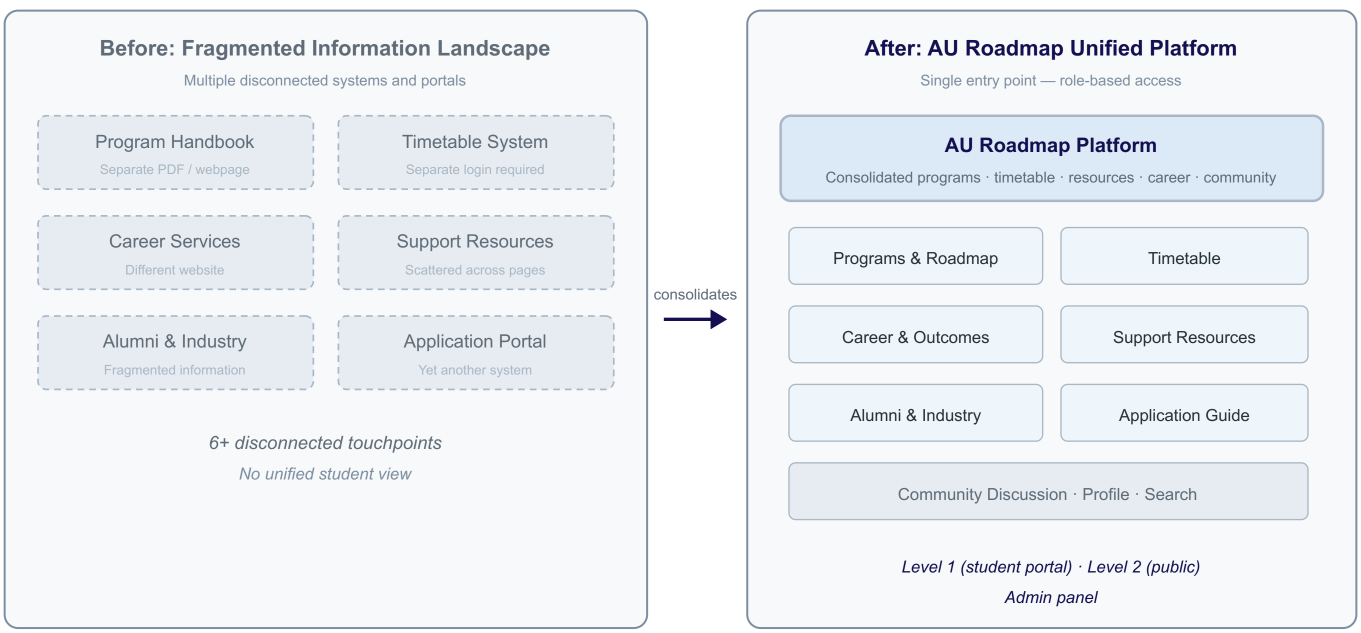

Figure 2. Fragmented information landscape versus the AU Roadmap consolidated platform.

4. Stakeholder Identification

4.1 Stakeholder Categories

The stakeholders for the AU Roadmap project fall into three broad categories: primary users, secondary stakeholders, and governance and delivery stakeholders.

4.2 Primary Users

Current Students are the primary users of Level 1, the authenticated student portal. They view their program roadmap, access timetable, browse resources, connect with community and industry content, and manage their profile.

Prospective Students are the primary users of Level 2, the public-facing exploration website. They browse programs, compare options, understand career outcomes, explore campus life, and access application guidance.

Administrators are the primary users of the admin panel. They create and update program and course records, manage alumni and industry data, maintain career information, and oversee user accounts and roles.

4.3 Secondary Stakeholders

Alumni contribute content and benefit from continued visibility. Industry partners contribute content and partnership visibility. Academic and support staff benefit indirectly from better-informed students. University marketing and recruitment teams benefit from a stronger public-facing program exploration experience.

4.4 Governance and Delivery Stakeholders

The academic mentor provides guidance, evaluation, and quality oversight. The university assigns the project and evaluates its deliverables. The project team is responsible for analysis, design, implementation, testing, and documentation.

4.5 Stakeholder Summary Table

Stakeholder	Category	Main Interest in the System
Current Students	Primary User	Personalised program roadmap and support access
Prospective Students	Primary User	Program discovery, comparison, and decision support
Administrators	Primary User	Efficient content and data management
Alumni	Secondary	Visibility and continued connection to the university
Industry Partners	Secondary	Student engagement and partnership visibility

Academic / Support Staff	Secondary	Better-informed students
Marketing / Recruitment	Secondary	Improved public-facing program presentation
Academic Mentor	Governance	Quality, coherence, and academic standards
University	Governance	Project outcomes, deliverables, and evaluation
Project Team	Delivery	Analysis, design, implementation, testing, and documentation

Table 3. Stakeholder summary for the AU Roadmap platform.

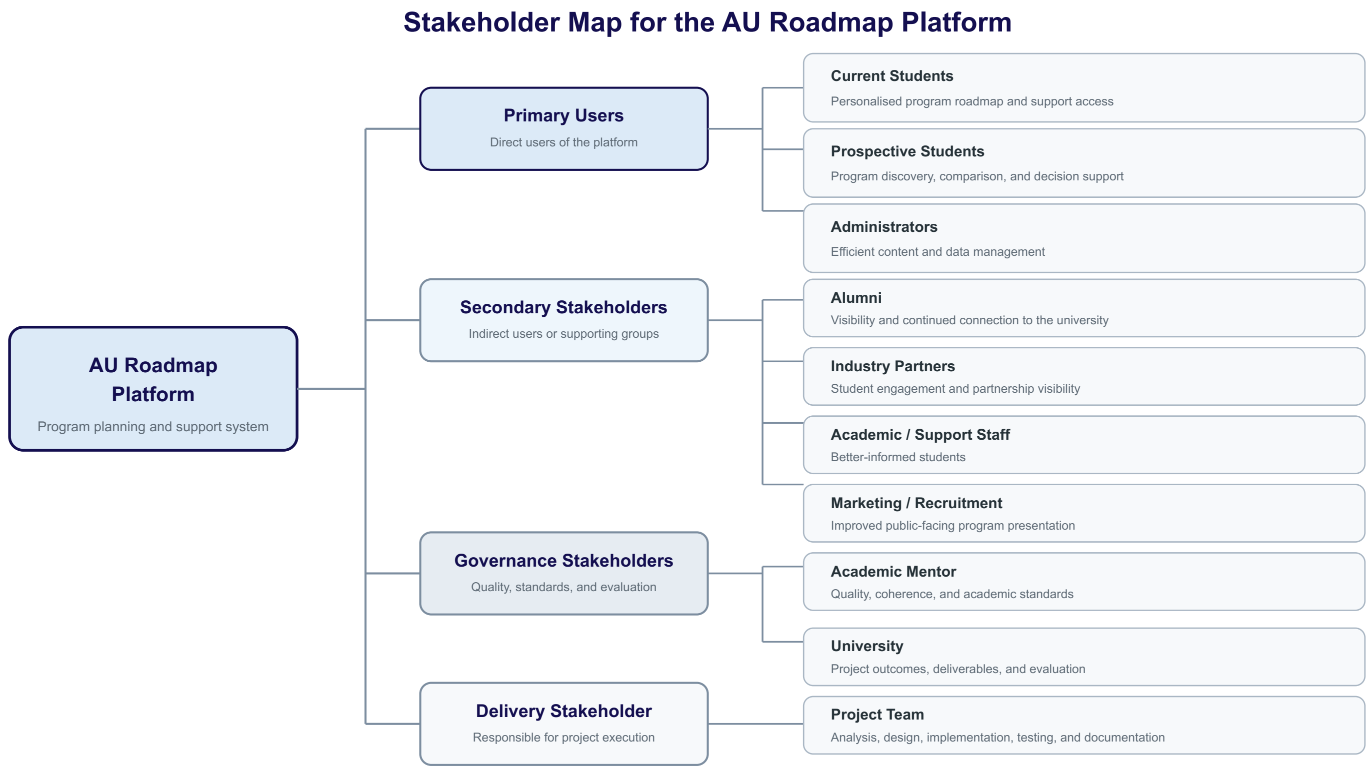

Figure 3. Stakeholder summary for the AU Roadmap platform.

5. User Profiles and Personas

5.1 User Profile: Current Student

Current students are enrolled in a degree program at Adelaide University. They range from first-year undergraduates to postgraduate students managing complex study and career decisions. They access the platform regularly throughout the semester and need quick, reliable access to their program structure, timetable, resources, and community connections.

Their main frustration is that this information currently requires multiple separate searches or system logins, and is not always presented in a structure that matches how they think about their academic progression.

5.2 User Profile: Prospective Student

Prospective students are individuals considering enrolment at Adelaide University. They may be domestic or international, school leavers or mature-age applicants. They access the platform from the public web without logging in. Their main frustration is that program and outcome information is often presented differently across university pages, making comparison difficult.

5.3 User Profile: Administrator

Administrators are authorized staff members responsible for maintaining the platform's structured content. They manage program records, course records, alumni profiles, industry partner information, career data, and user accounts. Their main frustration in a non-integrated environment is the duplication of effort across separate systems.

6. User Environment and Context of Use

6.1 Physical and Technical Environment

Users access the AU Roadmap platform through standard web browsers on desktop and mobile devices. The platform is currently implemented as a local development prototype using Node.js with Express (backend), Vue 3 and Vite (frontend), and MySQL (database).

6.2 Social and Organisational Environment

The platform sits within the broader Adelaide University digital ecosystem. Current students expect a task-oriented experience; prospective students expect a polished, informative exploration experience; administrators expect a predictable, efficient management interface.

6.3 Usage Patterns

Current students access the portal regularly for quick checks. Prospective students visit less frequently but for longer exploration sessions. Administrators access the admin panel as needed for content updates.

7. User Needs and Expectations

7.1 Current Student Needs

Current students need a logical, structured view of their program roadmap showing which courses to take and when. They need to distinguish clearly between core and elective requirements. They need quick access to timetable information connected to their enrolled courses. They need a single entry point for support resources, study help, and wellbeing information rather than searching across separate university pages. They need to see industry and alumni connections relevant to their program. They need a discussion space to exchange questions and advice with peers. They need their profile page to display trusted university account information.

7.2 Prospective Student Needs

Prospective students need to explore available programs with clear descriptions, study structures, and entry requirements. They need to compare programs meaningfully. They need to see career outcomes data for each program. They need to understand campus life and the broader student experience. They need a clear application guide showing the steps toward enrolment.

7.3 Administrator Needs

Administrators need to manage program and course records through structured forms rather than raw queries. They need newly created courses to link correctly to relevant programs. They need to maintain alumni, industry, and career data efficiently. They need role management and user administration capabilities.

7.4 Quality Expectations

Across all user groups, four quality expectations emerge consistently. First, clarity: the information must be easy to read and understand. Second, structure: navigation must feel logical and task-oriented. Third, credibility: the platform must feel trustworthy, particularly where university account information is displayed. Fourth, access control: users expect that student-only content is protected and administrative functions are restricted.

8. Formal User Requirement Statements

The following section presents the core user requirements for the AU Roadmap Platform. These requirements are written in a formal requirement statement format using “The system shall…” to describe the expected system capabilities for each major user group. Each requirement is assigned a unique identifier to support traceability in later design, implementation, and testing stages.

8.1 Current Student Requirements

Req. ID	Requirement Statement
CUR-01	The system shall allow current students to view their program roadmap, including the recommended order and structure of their study progression.
CUR-02	The system shall allow current students to distinguish between core courses and elective courses within their program structure.
CUR-03	The system shall allow current students to view timetable information relevant to their enrolled courses and weekly academic schedule.

CUR-04	The system shall provide current students with access to study resources and support resources from a centralised student portal.
CUR-05	The system shall allow current students to access industry-related and alumni-related information to support career development and professional planning.
CUR-06	The system shall provide current students with access to a community discussion space where they can ask questions and share advice with peers and alumni.
CUR-07	The system shall display trusted university account information on the current student profile page, including institution-controlled personal and enrolment details where applicable.
CUR-08	The system shall restrict access to current-student-only pages to authenticated users with a current student role.

Table 4. Current student requirement statements (CUR series).

8.2 Prospective Student Requirements

Req. ID	Requirement Statement
PRO-01	The system shall allow prospective students to explore available programs offered by the university.
PRO-02	The system shall allow prospective students to view detailed program information, including program descriptions, study structure, and relevant academic information.
PRO-03	The system shall provide prospective students with career outcomes information associated with each program.
PRO-04	The system shall provide prospective students with access to campus life information to support their understanding of the broader student experience.
PRO-05	The system shall provide prospective students with a clear application guide that explains the main steps toward enrolment.
PRO-06	The system shall allow prospective students to access all public-facing exploration pages without requiring authentication.
PRO-07	The system shall prevent prospective student accounts from accessing authenticated student portal pages.

Table 5. Prospective student requirement statements (PRO series).

8.3 Administrator Requirements

Req. ID	Requirement Statement
ADM-01	The system shall allow administrators to create, view, update, and delete program records.
ADM-02	The system shall allow administrators to create, view, update, and delete course records.
ADM-03	The system shall allow administrators to link courses to relevant programs with roadmap metadata, including year level, semester, core/elective designation, and sort order.
ADM-04	The system shall allow administrators to manage alumni-related information displayed on the platform.
ADM-05	The system shall allow administrators to manage industry-related information displayed on the platform.
ADM-06	The system shall allow administrators to manage career outcomes content associated with programs.
ADM-07	The system shall allow administrators to manage user accounts and role assignments.
ADM-08	The system shall restrict administrative functions to authenticated users holding the administrator role.

Table 6. Administrator requirement statements (ADM series).

8.4 General Cross-Cutting Requirements

Req. ID	Requirement Statement
GEN-01	The system shall provide role-based access control distinguishing current students, prospective students, alumni, and administrators.
GEN-02	The system shall provide a navigation structure that presents relevant platform sections according to the authenticated user's role.
GEN-03	The system shall maintain a consistent visual interface and interaction pattern across public pages, student portal pages, and administrator pages.
GEN-04	The system shall protect restricted student and administrator pages from unauthorised access, returning a controlled denial response.
GEN-05	The system shall maintain accurate and up-to-date program, course, roadmap, and career-related information through administrator-maintained structured data.

Table 7. General cross-cutting requirement statements (GEN series).

9. Functional Requirements

The functional requirements below are written at system level and grouped by major system area.

9.1 Public Website and Discovery

ID	Requirement	User Group	Priority	Acceptance Logic
FR-01	The system shall provide a public homepage introducing the platform and directing users to major content areas.	Prospective students, general visitors	High	User can access homepage without authentication and navigate to key areas.
FR-02	The system shall provide a public program exploration page with search and filter capabilities.	Prospective students	High	User can browse and narrow programs by available criteria.
FR-03	The system shall provide individual program detail pages.	Prospective students	High	User can open a program page and view program-specific information.
FR-04	The system shall provide public career outcomes content.	Prospective students	High	User can access career outcomes content without authentication.
FR-05	The system shall provide public campus life content.	Prospective students	Medium	User can access student experience information without authentication.
FR-06	The system shall provide public application guidance content.	Prospective students	High	User can access application steps and guidance.
FR-07	The system shall provide public research and innovation pages as supporting university information.	Public visitors	Medium	User can browse research-related pages from the public area.

Table 8. Functional requirements for the public website and discovery layer (FR-01 to FR-07).

9.2 Authentication and Access Control

ID	Requirement	User Group	Priority	Acceptance Logic
FR-08	The system shall allow users to register and log in securely.	Students, prospective students	High	Valid users can register or log in and receive authenticated access where appropriate.

FR-09	The system shall distinguish user roles including student, prospective, admin, and alumni where applicable.	All authenticated users	High	System behaviour differs correctly by role.
FR-10	The system shall restrict student portal pages to authenticated student-related roles.	Students, alumni	High	Unauthenticated users are redirected or denied.
FR-11	The system shall restrict admin pages to authenticated administrators.	Admins	High	Non-admin users cannot access admin pages.

Table 9. Functional requirements for authentication and access control (FR-08 to FR-11).

9.3 Student Portal

ID	Requirement	User Group	Priority	Acceptance Logic
FR-12	The system shall provide a student dashboard summarising the student's context and key next actions.	Current students	High	Student can access dashboard after login.
FR-13	The system shall provide a roadmap page showing program structure by year and semester.	Current students	High	Student can view program sequencing in a structured layout.
FR-14	The system shall distinguish core and elective courses clearly on the roadmap.	Current students	High	Course type is visibly identified.
FR-15	The system shall provide a timetable page for schedule viewing.	Current students	Medium	Student can view schedule information in timetable format.
FR-16	The system shall provide a student industry connections page.	Current students	Medium	Student can browse industry-related content.
FR-17	The system shall provide access to grouped student resources.	Current students	High	Student can access support categories from one place.
FR-18	The system shall provide a community discussion area with threads and replies.	Current students, alumni	Medium	Authenticated users can read and contribute to discussions.

FR-19	The system shall support image upload for community posts.	Current students, alumni	Medium	User can attach and display an image in a discussion post.
FR-20	The system shall provide a profile page showing trusted account information and avatar management.	Current students	Medium	Student can view account details and update avatar.

Table 10. Functional requirements for the student portal (FR-12 to FR-20).

9.4 Administrative Functions

ID	Requirement	User Group	Priority	Acceptance Logic
FR-21	The system shall allow administrators to manage programs.	Admins	High	Admin can create, update, view, and delete program records where supported.
FR-22	The system shall allow administrators to manage courses.	Admins	High	Admin can create, update, view, and delete course records where supported.
FR-23	The system shall allow administrators to link courses to programs with roadmap metadata.	Admins	High	Admin-created course-program relationships are stored correctly.
FR-24	The system shall allow administrators to manage alumni, industry, and career data.	Admins	High	Admin can access and update these content domains.
FR-25	The system shall allow administrators to manage users and roles.	Admins	High	Admin can access user management and role-related operations.
FR-26	The system shall provide an admin dashboard that summarises management areas.	Admins	Medium	Admin can access overview information after login.

Table 11. Functional requirements for administrative functions (FR-21 to FR-26).

9.5 Search, Routing, and Validation

ID	Requirement	User Group	Priority	Acceptance Logic
FR-27	The system shall provide unified	Public and	Medium	Search returns relevant

	search across relevant content domains.	authenticated users		structured results.
FR-28	The system shall support direct routing to public, student, and admin pages according to access rules.	All users	High	Navigation and route protection behave correctly.
FR-29	The system shall validate required user input before database actions are performed.	All users	High	Invalid requests return controlled validation errors.
FR-30	The system shall provide appropriate authorisation error responses for protected routes.	All users	High	Unauthorised requests return the expected denial response.

Table 12. Functional requirements for search, routing, and validation (FR-27 to FR-30).

10. Non-functional Requirements

Quality-related requirements that shape the user experience and project viability are specified below.

10.1 Usability

ID	Requirement	Rationale	Priority
NFR-01	The system should use clear and understandable navigation labels.	Users must find relevant information quickly.	High
NFR-02	The system should organise pages according to user tasks rather than internal data ownership alone.	Information should feel natural to users.	High
NFR-03	The system should maintain consistent layouts and interaction patterns across related sections.	Consistency reduces cognitive load.	High
NFR-04	The system should reduce unnecessary clicks for common tasks such as roadmap viewing, program browsing, and support access.	Efficiency supports repeated use.	Medium

Table 13. Non-functional requirements: usability (NFR-01 to NFR-04).

10.2 Accessibility

ID	Requirement	Rationale	Priority
NFR-05	The system should use readable typography, spacing, and content hierarchy.	Supports readability and comprehension.	High
NFR-06	The system should use sufficient visual contrast for major interface elements, in alignment with WCAG 2.2 [4].	Supports visual accessibility.	High
NFR-07	The system should avoid cluttered layouts and overwhelming text blocks where possible.	Supports comprehension for diverse users.	Medium

Table 14. Non-functional requirements: accessibility (NFR-05 to NFR-07).

10.3 Performance

ID	Requirement	Rationale	Priority
NFR-08	The system should load key pages and return common API responses efficiently in the local development environment.	Performance affects trust and usability.	High
NFR-09	The system should remain stable under lightweight repeated public API requests used in prototype demonstrations.	Supports demo readiness and technical credibility.	Medium

Table 15. Non-functional requirements: performance (NFR-08 to NFR-09).

10.4 Security

ID	Requirement	Rationale	Priority
NFR-10	The system should protect authenticated routes using token-based access control.	Prevents unauthorised access.	High
NFR-11	The system should enforce role-based restrictions between student, public, and admin areas.	Protects sensitive functions and content.	High
NFR-12	The system should validate input and return controlled validation errors.	Reduces invalid writes and unstable behaviour.	High

Table 16. Non-functional requirements: security (NFR-10 to NFR-12).

10.5 Maintainability

ID	Requirement	Rationale	Priority
NFR-13	The system should maintain modular separation between frontend pages, backend controllers, and database structures.	Supports iterative maintenance.	High
NFR-14	The system should support future expansion of programs, pages, and content categories.	The prototype should be extendable.	Medium

Table 17. Non-functional requirements: maintainability (NFR-13 to NFR-14).

10.6 Feasibility

ID	Requirement	Rationale	Priority
NFR-15	The solution should remain realistic for a prototype developed within course project constraints.	Scope control is necessary for delivery.	High
NFR-16	The implementation roadmap should remain achievable using the selected web stack and available team resources.	Supports delivery confidence.	High

Table 18. Non-functional requirements: feasibility (NFR-15 to NFR-16).

11. External Interface Requirements

The external interfaces relevant to the platform are summarised below.

The AU Roadmap prototype interacts with several external systems and services at the boundary level. The frontend communicates with the backend exclusively through RESTful JSON APIs over HTTP. The backend connects to a local MySQL database for all persistent data operations. Cloudinary is integrated for cloud-based image storage, supporting avatar upload and community post image upload. Cloudflare Turnstile is used on the registration and login pages for bot protection and CAPTCHA-based validation.

No direct integration with live Adelaide University student information systems, enrolment databases, or timetable scheduling systems is included in the current prototype scope. These would form part of any future production extension.

12. Assumptions, Dependencies, and Constraints

The assumptions, dependencies, and constraints that shaped the requirement analysis are discussed below.

12.1 Assumptions

Users have basic web literacy and can navigate a standard website without training. The platform will operate in modern browsers with JavaScript enabled. Student authentication will use email and password rather than university single sign-on in the prototype phase. The platform will be a supplementary tool, not a replacement for official university systems such as the LMS or student enrolment portal. Content accuracy depends on administrator maintenance.

12.2 Dependencies

The requirement set depends on the two-level platform architecture defined in the project proposal. The prototype's functional scope depends on the availability of seed data for demonstration. Admin functions depend on the relational database schema linking programs, courses, and other content entities. Evaluation of the platform against requirement criteria depends on testing evidence collected during the testing phase.

12.3 Constraints

The project is constrained by the 10-week academic semester timeline, the available team resources, the requirement to use a specific technology stack (Vue 3 frontend, Express backend, MySQL database), and the scope expectations of a course-based prototype rather than an enterprise production system. These constraints are reflected in NFR-15 and NFR-16.

13. Requirement Prioritisation

Functional and non-functional requirements are assigned one of three priority levels: High, Medium, or Low. High-priority requirements are considered essential for the prototype to demonstrate its core value. Medium-priority requirements add meaningful value but the prototype would remain evaluable without them. Low-priority requirements are desirable enhancements deferred to future work.

Priority assignments were based on: directness of relationship to the core problem statement; importance to the primary user groups; contribution to the evaluation criteria defined in the project proposal; and feasibility within the project timeline and resources.

14. Requirement Traceability to the Prototype

The mapping below links the requirement analysis to the implemented prototype structure, showing that the major functional areas are addressed by corresponding prototype pages and backend modules.

Requirement Domain	Prototype Pages / Modules	Covered Reqs
Public discovery and information	Homepage, Programs, Program Detail, Career, Campus, Application, Research	FR-01 to FR-07; PRO-01 to PRO-07
Authentication and access	Login, Register, Auth controllers,	FR-08 to FR-11; GEN-01, GEN-

control	Route guards	02, GEN-04
Student portal and roadmap	Dashboard, Roadmap, Timetable, Profile	FR-12 to FR-15, FR-20; CUR-01 to CUR-04, CUR-07, CUR-08
Student resources and community	Resources, Community, Industry Connections	FR-16 to FR-19; CUR-05, CUR-06
Administrative content management	Admin Dashboard, Program Mgmt, Course Mgmt, Alumni Mgmt, Industry Mgmt, Career Mgmt, User Mgmt	FR-21 to FR-26; ADM-01 to ADM-08; GEN-05
Search, routing, validation	Search, Router config, Middleware, Error handlers	FR-27 to FR-30; GEN-03, GEN-04

Table 19. Requirement-to-prototype traceability mapping.

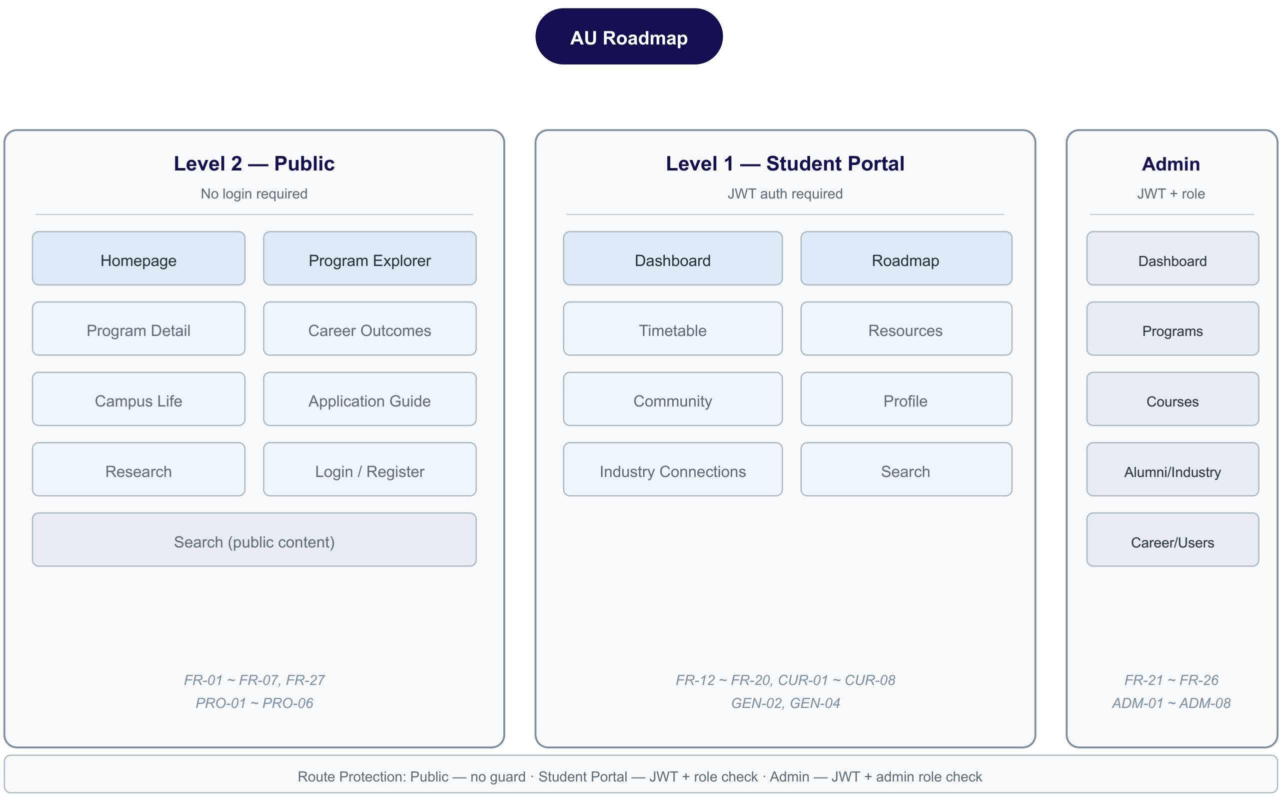

Figure 4. Platform sitemap showing public, student portal, and administrator page routes.

15. Evaluation Against User-Centred Project Goals

The requirement analysis aligns strongly with the project evaluation criteria across all major dimensions.

15.1 Accessibility

The requirements emphasise readable content, clear structure, appropriate spacing, and understandable navigation. These support accessibility at both the interface and information level. NFR-05 through NFR-07 explicitly address typography, contrast, and layout considerations.

15.2 Usability

The requirements focus on easy navigation, reduced information fragmentation, and stronger task-based organisation. NFR-01 through NFR-04 directly support usability expectations through clear labels, task-oriented page organisation, consistent interaction patterns, and reduced unnecessary clicks.

15.3 Content Quality

The requirements assume that content must be understandable, accurate, and relevant to user tasks. This is reflected in the user needs analysis throughout Sections 4–8. The admin requirements (FR-21 through FR-26) further support content quality by enabling structured maintenance of program, course, alumni, and career data.

15.4 Information Structure

The separation of public, student, and admin pathways is one of the most direct outcomes of the requirement analysis. The user profiles in Section 6, the distinct user needs in Section 7, and the role-based access requirements (FR-09 through FR-11) all converge on three distinct information environments: public exploration, authenticated student support, and administrative management.

15.5 Technical Performance

NFR-08 and NFR-09 address the performance evaluation criterion. Testing evidence gathered during the prototype evaluation phase confirms that the implemented system meets these expectations: public APIs remained responsive under light and moderate local test loads with a 0% error rate.

15.6 Feasibility

NFR-15 and NFR-16 explicitly constrain the requirement scope to what is realistic for a course-based prototype. All thirty functional requirements are supported by working prototype pages, as shown in the traceability mapping in Section 14.

15.7 Completeness

The requirement set spans the major functional domains of the system — public discovery, authentication, student planning and community, administrator content management, search, and route-level security. The cross-references in Section 2.3 document how each subsequent deliverable builds on this requirement analysis.

16. Requirement Summary

The AU Roadmap project addresses a user-centred problem grounded in the actual information-seeking behaviour of university audiences: program-related information is not always easy to find, compare, or connect across the student journey. Different user groups need different information environments, but all depend on clear structure, understandable content, and reliable access.

Beyond restating the user profiles, the requirement analysis reveals three structural insights. First, the roadmap is not just a feature but the organising metaphor for the entire Level 1 student experience — dashboard, timetable, resources, and community all gain their relevance from relationship to roadmap-oriented planning. Second, the public and authenticated environments need to feel integrated in brand and information quality but separated in access and task design. Third, administrator tools need to remain domain-oriented rather than mixed with user-facing journeys, confirming the value of the third administrative environment.

These findings justify the two-level architecture, the separation of public and authenticated journeys, the centrality of roadmap-oriented planning, and the inclusion of supporting admin tools. They also provide a strong foundation for the final report, implementation roadmap, and system evaluation.

References

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[4] Web Content Accessibility Guidelines (WCAG) 2.2. World Wide Web Consortium (W3C) Recommendation, 2023.

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Appendix — Condensed Requirement Summary

A condensed, business-level view of the requirement set is presented below, suitable for inclusion in summary deliverables.

A. Core Business Requirements

The platform must support current students and prospective students through different but connected journeys, present program information in a more structured and user-centred way than fragmented standalone pages, and connect academic planning with broader student support and employability-related context.

B. Core User Requirements

Current students must be able to view and understand their program roadmap. Prospective students must be able to explore and compare programs easily. Administrators must be able to maintain core program and course information.

C. Core Quality Requirements

The system must be usable, structured, and credible. It must support secure access control, and it must remain feasible within the defined project scope.